

# Do's, Don'ts, and Responsibilities<sup>1</sup>

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**Learn the mathematics, not just how to finish the page.** Use AI to ask better questions and work through confusion, not replace the thinking you are here to practice. Training for a marathon on a motorcycle misses the point.

## Which rules apply?

Follow institutional rules, approved accommodations, and assignment-specific instructions. Part II does not override them. When permission is unclear, ask your instructor before proceeding.

## Allowed under these defaults

- **Study support:** explanations of concepts, background review, extra practice questions, and exploration beyond the syllabus. You do not need to be stuck first.
- **Limited assignment help:** after a meaningful attempt, request a small hint or feedback locating an error—not the missing argument or a replacement solution. Do the next step yourself.
- **Mechanical help:** troubleshoot L<sup>A</sup>T<sub>E</sub>X or identify typos without adding or rewriting mathematics.

## Not allowed unless the instructor explicitly authorizes it

- Use AI on an exam, quiz, or other task designated as independent or no-AI work.
- Ask AI to start an assigned problem, supply its solution or key argument, or assemble the solution through successive hints. Paraphrasing generated work does not make it your own.
- Have AI write or substantively rewrite your submitted proof, explanation, or code. **Writing is part of doing the mathematics.** Broader AI collaboration on assigned work requires explicit permission.

## Your responsibilities

**Verify and understand.** Check mathematics and references; be able to justify each submitted step from permitted definitions and results. Understanding something later does not retroactively authorize prohibited assistance.

**Disclose relevant use.** For submitted work, identify the tool, its contribution, and your checks; follow additional course requirements. Keep relevant records, not unrelated private conversations.

**Protect materials.** Do not upload restricted assessments, others' work, personal information, or unpublished material without authorization. AI-use permission is not automatically permission to upload course materials.

**Handle surprises openly.** If AI unexpectedly exceeds the permitted scope, stop that branch, keep the relevant exchange, and ask your instructor how to proceed. Do not silently incorporate it.

## Before you submit

**Permission:** Was this kind of help allowed? **Ownership:** Did I do the required thinking and writing?

**Verification:** Can I justify the work? **Transparency:** Have I accurately described the assistance?

<sup>1</sup>This note is loosely based on Abbey Bourdon's ([bourdoam@wfu.edu](mailto:bourdoam@wfu.edu)) original document, which was adapted from Pavel Etingof's "Use of AI in mathematical research: A guide for young mathematicians." The PDF was created with the help of ChatGPT 6 Pro. For questions and comments, contact Fan Yang at [yangf@wfu.edu](mailto:yangf@wfu.edu).

## PART II

## Learn Mathematics with AI

Use AI to untangle a definition, get unstuck, or follow a mathematical curiosity beyond your current course. Start with the section you need. For coursework, the rules in Part I still apply.

### First: check the math, not the confidence

**AI can be wrong—even when it says your proof is wrong.** Look for reasons you can check, not just a confident verdict.

- **The question:** Are the objects, assumptions, and quantifiers unchanged?
- **The reasoning:** Can you justify the key step? Check a calculation or a small case yourself.
- **The tools:** Do the cited results apply? Are you using the very result you are trying to prove?
- **The examples:** Do they satisfy every required condition?
- **The sources:** Open the actual reference and check what it says.

“Please check carefully” is not a verification method. Agreement between AI responses is not a substitute for proof.

### Ask for an assessment, not agreement

AI can sometimes follow your opinion rather than assess the mathematics independently.<sup>2</sup> When checking a claim or argument, avoid building your preferred verdict into the question.

**Avoid:** “This proof is correct. Explain why.”

**Try:** “Does this argument establish the claim? Identify any invalid or unjustified steps and explain your assessment. Do not rewrite the proof.”

Share your reasoning, not a required conclusion. “Find why my proof is wrong” also presupposes a verdict. If you disagree, give a mathematical reason and ask it to reassess. Neutral wording helps frame the task; it does not guarantee a correct answer.

### 1. Tell the AI where you are starting

Give your **level, relevant prior knowledge, and learning goal**. “I know matrix calculations but am new to proofs” is more useful than a course title alone. Say which definitions and notation you are using when they matter.

#### A reusable opening—replace the colored fields

I am studying [topic] at [level]. I am comfortable with [background], but new to [unfamiliar material]. I want to understand [goal]. Please give [a short explanation / a question / a hint]. Ask first if missing background would change your explanation.

**For coursework, add:** We have covered [definitions and results], but not [later material]. This task permits [tools and AI help].

**What you know, what a task permits, and what you want to learn are different.** Restrict tools when practicing a particular method; welcome clearly labeled new ideas when exploring. Save and update your background note instead of expecting the AI to infer it.

<sup>2</sup>For research on this tendency, see Sharma et al., [Towards Understanding Sycophancy in Language Models](#) (2023).

**UNDERSTAND AN IDEA • TRY IT YOURSELF****2. Take a definition apart—then test it**

A dense definition can hide several separate questions. What are the objects? What extra structure is supplied? Which conditions must hold? What is this definition trying to capture? Ask for those pieces before asking for a simpler version of the whole paragraph.

**Unpack the concept**

Break [definition] into its objects, structure, and conditions. Explain the motivation using [familiar ideas], then connect the intuition to the exact definition. Work through one concrete example. If you use an analogy, explain where it stops working.

**Next, test the boundaries.** Ask for a few examples, non-examples, and near-misses without their labels. Classify them from the definition, then compare your reasoning with the AI's. Try constructing your own example, too; checking its properties is part of the exercise.

**Still confused? Show your current understanding.** "I think this means [your explanation]; ask one question to check it." A targeted question can reveal whether you need one missing prerequisite or a different explanation. Identify the first symbol or step you cannot explain, rather than restarting the whole subject.

**Need practice? Name the skill, not just the topic**

"Three problems on derivatives" could mean computation, interpretation, or proof. Specify the skill you want to practice and the background you can use.

Give me three extra questions on [topic], using [background], to practice [skill]. Include a direct application, a method-choice question, and a variation. Show one at a time, without answers or method labels. Wait for my attempt before giving feedback.

After feedback, try a different-looking problem using the same idea, without hints. **Check the question, too:** generated exercises and answer keys can be wrong. "Prove that" does not make a statement true. Do not use a request for a "similar example" to obtain a disguised solution to restricted homework.

**3. Get one hint—then take back the problem**

When the task is to develop independent problem-solving, make a meaningful attempt first: write the definition, test a small case, draw a picture, or locate where your approach breaks down. Bring your work, not just the question.

Here is my attempt: [work]. I am stuck at [step]. Give one small hint about that obstacle, not a solution or a plan for the whole proof. Then wait while I try it.

Put the chat aside and act on the hint. If it assumes something unfamiliar, ask about that idea instead of requesting each next step. **A solution can be outsourced one hint at a time.** You should still be choosing the arguments and writing the proof.

You do not have to struggle before asking what a new definition means. Reading a worked example while learning a new topic is different from outsourcing an exercise intended for independent work.

## READ THE ARGUMENT • FOLLOW THE CONNECTIONS

## 4. Read a proof for its plan, not just its lines

Following every line is not the same as seeing why the proof takes that route. For a proof in a textbook, lecture notes, or an article, ask how the parts fit together, then focus on the transitions you cannot justify. Provide the actual passage when sharing it is permitted; a theorem number alone is not enough.

**See the structure**

Here is the proof: [passage]. Identify its main idea and intermediate goals. Explain [difficult transition] using [definitions and known results]. Do not silently substitute a different proof. Distinguish intuition from a complete argument.

Then close the explanation and sketch the plan in your own words: what was constructed, what was estimated, and what finished the argument? Reopen the source to check, not to copy. Comparing two available proofs can also reveal what each method makes easier to see.

**Ask what the assumptions are doing**

Choose one hypothesis. Where does the proof use it? Could the conclusion survive without it? What about the converse? Do not assume every stated hypothesis is necessary.

What happens if we remove or weaken [hypothesis]? Distinguish “this proof no longer works” from “the conclusion is false.” Suggest a case for me to investigate, without assuming which outcome is correct.

A proposed counterexample must satisfy all the remaining assumptions and fail the conclusion. If the original proof breaks, that alone does not rule out another proof.

**Feedback on your own proof.** When allowed, ask the AI to quote the exact step it questions and distinguish a false claim, a missing justification, and a stylistic suggestion. Check the criticism and make any repair yourself; do not hand over the writing.

## 5. Follow a familiar idea into a new setting

Ask “Where have I seen this structure before?” Look for a precise connection, not just a shared symbol.

Connect [new concept] to [familiar concept]. What structure is shared? What changes? Distinguish exact relationships from analogies, and flag new prerequisites. Give me one small task to test the connection.

**Follow the + sign.** Vector addition gives an abelian group; scalar multiplication adds more structure. In the rational Cauchy-sequence construction of  $\mathbb{R}$ , addition is defined by

$$[(a_n)] + [(b_n)] = [(a_n + b_n)].$$

Once you know Cauchy sequences and equivalence classes, ask why the sum is Cauchy and why its equivalence class does not depend on the chosen representatives. This connects checking properties of an operation with constructing an operation that is well-defined.

**Connect ideas within a subject, too.** Pick a definition, two examples, and a theorem. Explain their links yourself, then ask the AI to question or extend them. Label the links: “is an example of,” “implies,” or “is equivalent under these assumptions.” A small map you can explain beats a large one you can only admire.

## GO BEYOND THE SYLLABUS • KEEP THE LEARNING

## 6. Learn a related topic your course has not reached

Curiosity is a good reason to start. You can ask AI to help you learn a new topic, not just answer questions about material already taught. Start with something you know and a question you would like to understand.

1. **Pick a destination.** Replace “Teach me probability” with a concrete goal. For example: “I know matrices and eigenvectors and want to understand the long-term behavior of a simple Markov chain.”
2. **Ask for a short route.** Request a few milestones, separating essential prerequisites from optional extensions. Start with the first milestone; do not collect an entire generated course before doing any mathematics.
3. **Learn one piece and use it.** Ask for motivation, an exact definition, and a worked example. Then calculate another example, explain a distinction, or prove a small claim yourself. Fill a missing prerequisite and return to the main question.
4. **Check against a real source.** Find one accessible textbook or set of notes, verify the reference, and read the relevant section. Finish at a concrete milestone you can demonstrate, not when the chat declares you an expert.

### Start a small learning project

I know [background] and want to learn [new topic] to understand [question or goal]. Suggest a short route, separating essential prerequisites from optional extensions. Teach one step at a time, with a concrete example and a small task for me. Recommend one accessible reference I can check.

For the Markov-chain example, a first milestone could be interpreting a transition matrix, computing a few steps, and explaining what a stationary distribution means. A later question is whether repeated transitions approach it, and under what conditions. You need not learn all of probability before investigating this small example.

## 7. Use pictures and code to test a prediction

**Predict** → **experiment** → **explain** → **prove**. Ask AI to help build a small plot or computation, not just produce an impressive picture. Before plotting  $x^n$  on  $[0, 1]$ , predict its behavior at a fixed point in  $(0, 1)$ , at 1, and near 1.

Explain what the code tests, check an output by hand, and consider sampling and numerical limits. Pictures and successful tests are not proofs. For coursework, follow the code-generation rules; if implementing the method is the exercise, request a debugging hint instead.

### Before you close the chat

**Try it alone.** Explain the idea or solve a fresh problem without hints. Revisit it another day, mixed with other topics so the method is not announced.

**Keep one next step.** Write down one thing you can now do and one question to return to. If answers keep changing without reasons, stop collecting prompts: take the exact question and your attempt to a teacher, tutor, or study partner.

## References and sources

- [1] STEM Education: Shaping Future Learning Practices in the Age of AI, Springer Nature Singapore, June 2026.
- [2] Brayan Díaz and Miguel Nussbaum. Artificial intelligence for teaching and learning in schools: The need for pedagogical intelligence. *Computers & Education*, 217 (2024) 105071.
- [3] Pavel Etingof, Use of AI in mathematical research: A guide for young mathematicians.
- [4] James M. Hyman, Learning with AI, public companion materials: [Study Partner Protocol](#) and [Define Personal Preferences](#). Background, course scope, learning goals, and study conversations.
- [5] James M. Hyman, [The departmental adoption kit](#). A reference for the proposed policy structure in Part I: explicit permission, assignment-specific boundaries, disclosure, and verification. This draft does not adopt the kit wholesale.
- [6] James M. Hyman, [Spiral problems](#). Examples of organizing a task around an attempt, feedback, independent checking, and a new question.
- [7] Cat Hicks, [Learning Opportunities](#). Learning exercises originally designed for AI-assisted coding, including prediction, recall, and spaced review. Mathematical applications here are adaptations.
- [8] Qi Guo, [Evaluating mathematical merit](#), Proofs and Prompts, August 27, 2026. See the discussion of education and mathematical knowledge networks.
- [9] Edgar Costa, [How to Tame Your LLM: Harnesses for Rigorous Work](#), September 4, 2026. Research-oriented material on keeping records, checking outputs, and setting stopping conditions.
- [10] Katherine E. Stange, [Do Androids Dream of Electric Sheaves?](#) An annotated collection of resources on mathematics and AI, through which the additional materials above were located.

Web resources consulted September 17, 2026.